Question Paper Code: 25901230

MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026
Fourth SEMESTER
Discipline Specific Elective (DSE) - MG4DSEECT201 - 3D PRINTING AND ITS APPLICATIONS
(2024 ADMISSION ONWARDS)

Duration: 2 hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I)and Appreciation(Ap)

ANSWER KEY

Part A
MCQ
Answer all questions
Each question carries **1** marks

1.	Recall the powder bed fusion process using a laser.	[U] / [CO1]
	c). SLS	
2.	What is a major advantage of parametric assemblies in CAD systems?	[U] / [CO2]
	a). They allow automatic updates of related components when onedimension changes	
3.	Orientation of parts in build preparation software mainly affects	[An] / [CO3]
	c). Build time and mechanical strength	
4.	Warping primarily occurs due to	[C] / [CO4]
	b). Uneven cooling and shrinkage	
5.	Classify CNC machining under manufacturing types.	[U] / [CO1]
	b). Subtractive process	
6.	State the goal of computational design.	[U] / [CO2]
	a). Automated performance based optimization	
7.	Customization in 3D printing is most beneficial for	[An] / [CO3]
	b). Complex and low-volume parts	
8.	Preventative maintenance reduces	[C] / [CO4]
	a). Downtime	
9.	Recognize an advantage of AM in aerospace sector.	[U] / [CO1]
	c). Complex lightweight structures	
10.	Name the software stage that prepares a model for printing.	[U] / [CO2]
	b). Slicing	
11.	While framing AM applications, which cost element is significantly reduced?	[An] / [CO3]
	c). Tooling cost	
12.	The controller board in a 3D printer kit is responsible for	[C] / [CO4]
	d). Interpreting G-code and controlling movements	

13. Classify post processing under AM stages. [U] / [CO1]

c). Post production stage

14. Name the analysis used to evaluate stress in a model. [U] / [CO2]

b). Finite Element Analysis

15. The concept of printing spare parts only when required is known as [An] / [CO3]

a). On-demand manufacturing

16. Which representation divides the model into small volumetric elements for AM? [An] / [CO3]

c). Voxel representation

[1x16 = 16]

Part B

Short Answer Questions
Answer any **6** questions
Each question carries **5** marks

17. Discuss the role of energy input and scan speed in controlling microstructureduring Directed Energy Deposition. [U] / [CO1]

- Energy Input (Laser/Beam Power):
  - Higher energy → deeper melt pool → slower cooling → coarser grains
  - Lower energy → rapid solidification → finer microstructure
  - Excess energy may cause porosity or distortion
- Scan Speed:
  - High scan speed → less heat input → fine grains, higher hardness
  - Low scan speed → more heat accumulation → coarse grains
- Combined Effect:
  - Determines cooling rate, grain size, and phase formation
  - Proper balance ensures desired mechanical properties

18. Define topology optimization. [U] / [CO2]

- Topology Optimization is a computational design method that:
  - Optimizes material layout within a given design space
  - Based on loads, constraints, and boundary conditions
- Objective: Maximize performance while minimizing material usage
- Widely used in lightweight and efficient structures

19. Distinguish between industrial and consumer applications in manufacturing. [An] / [CO3]

Aspect	Industrial Applications	Consumer Applications
Purpose	Mass production, heavy-duty	Personal use, customization
Scale	Large-scale	Small-scale
Materials	Metals, composites	Plastics, resins
Precision	Very high	Moderate
Cost	High investment	Low cost
Examples	Aerospace parts, automotive	Toys, phone cases

20. Explain the advantages of Binder Jetting. [U] / [CO1]

- No need for high thermal energy (no melting)
- Can print complex geometries
- Supports wide range of materials (metals, sand, ceramics)
- Fast printing speed
- No support structures required (powder acts as support)
- Suitable for mass production

21. Explain the advantages of extrusion based printing. [U] / [CO1]

- Low cost and easy to use
- Wide availability of materials (PLA, ABS)
- Minimal material wastage
- Suitable for rapid prototyping
- Safe operation (no lasers/chemicals)
- Easy maintenance

22. Explain the importance of digital workflow integration. [U] / [CO2]
- Ensures smooth transition from design to production
- Reduces human errors
- Enables automation and efficiency
- Allows real-time monitoring and control
- Improves repeatability and quality
- Supports data-driven manufacturing

23. Indicate the factors affecting the choice of tooling method. [An] / [CO3]
- Production volume
- Material type
- Cost constraints
- Required accuracy and surface finish
- Lead time
- Complexity of design
- Durability requirements

24. Comprehend the issue of layer shifting and describe its impact on print quality. [C] / [CO4]
- Definition:
  - Misalignment of layers during printing
- Causes:
  - Mechanical issues (belt looseness, motor errors)
  - Sudden movement or vibration
  - Software glitches
- Impact:
  - Distorted geometry
  - Poor dimensional accuracy
  - Weak structural integrity
  - Visible defects in surface finish

[5x6 = 30]

Part C
Essay Questions
Answer any 2 questions
Each question carries 12 marks

25. Explain the advantages of powder based printing. [U] / [CO1]
- No need for support structures (unsintered powder acts as support)
- Ability to produce highly complex geometries and internal features
- Suitable for wide range of materials (metals, polymers, ceramics)
- Enables high precision and fine resolution
- Reduced material wastage (unused powder can often be reused)
- Ideal for batch production of multiple parts simultaneously

26. Explain the importance of assembly modeling. [U] / [CO2]
- Helps visualize how multiple components fit together
- Detects interference or collision issues before manufacturing
- Improves design accuracy and functionality
- Allows simulation of movement and kinematics
- Facilitates easy modification and updates (parametric design)
- Enhances collaboration between design and manufacturing teams

27. Relate material properties to functional performance in AM components. [An] / [CO3]

- Strength & stiffness → Determines load-bearing capacity
- Thermal resistance → Important for high-temperature applications
- Density → Affects weight and efficiency (e.g., aerospace parts)
- Surface finish → Influences friction, wear, and aesthetics
- Anisotropy (layer-wise properties) → Affects strength in different directions
- Fatigue resistance → Important for cyclic loading conditions

Proper material selection ensures performance, durability, and reliability of AM components.

28. Design a 3D printed component while justifying choices related to wallthickness,infill density,and support structures. [C] / [CO4]

When designing a component:

(a) Wall Thickness

- Must be thick enough for strength but not excessive
- Thin walls → lightweight but may break
- Thick walls → stronger but increases material and print time
- Chosen based on load and material type

(b) Infill Density

- Low infill (10–30%) → lightweight, faster printing
- High infill (50–100%) → stronger, more durable
- Select based on functional requirement (strength vs weight)

(c) Support Structures

- Required for overhangs and complex geometries
- Ensures proper shape during printing
- Should be minimized to reduce material use and post-processing
- Orientation of part can reduce need for supports

[12x2 = 24]